Width and the Delay of Generalization on Modular Arithmetic: A Censoring-Aware, Falsifiable Test

Anonymous Author(s)
Affiliation
Address
email

Abstract

In grokking, a network fits its training set long before it generalizes. We ask how this delay depends on model width on modular addition ($p = 53$), and test a falsifiable account whose predictions (A1–A3) and decision rules were written down before the main analysis; the plan was amended once after partial results, and the negligibility bands used to label verdicts were fixed after seeing the results. All 156 distinct runs (single-layer transformers, plus an MLP and multiplication as robustness checks) are kept, including the ones that never generalize within the step budget, which we treat as right-censored. Under fixed learning rate, weight decay and initialization, wider transformers generalize sooner: the censoring-aware slope of $\log t_{\text{gen}}$ per doubling of width is -0.25 (95% CI $[-0.38, -0.10]$, widths 16–256, 5 seeds per width), but the interval's upper end sits at the edge of a negligible effect (-0.10) and restricted to widths ≥ 32 with a longer budget it includes zero (-0.31, $[-0.62, 0.12]$); in 100,000-step checks with 3 seeds, 1 of 3 runs at width 16 and 2 of 3 at width 32 generalize, late (35,000–58,000 steps), so the 30,000-step cells at narrow widths are lower bounds; the width-8 net reaches 0.56–0.97 validation accuracy in most runs without reaching 99% training accuracy, a separate regime that is not grokking. The inverse dependence of the delay on weight decay is in the predicted direction at both widths tested, with an exponent of -1.01 ($[-1.38, -0.66]$) at width 128 and a shallower, imprecise one at width 64. The proposed weight-norm mechanism is not supported: whether width acts only through the norm at fit depends on how the norm is measured (refuted with the total L_2 norm, which is collinear with width by construction, inconclusive with the per-parameter RMS norm), and a larger initial scale, which raises the norm at fit at width 128, shortens rather than lengthens the delay (direction-only, marginal); the intended norm-matched comparison could not be built, so mediation by the norm remains untested. With the learning rate scaled as $1/\sqrt{d}$ the width advantage is weaker (at widths 128 versus 32, 3/5 versus 2/5 runs generalize, against 5/5 versus 1/5 at fixed rate), but most scaled-rate runs are censored, so whether it depends on the learning rate stays open; and a critical-training-fraction shift with width is inconclusive. Results hold for one task family, one prime, a narrow width range and few seeds; we state where they are inconclusive.

1 Introduction

Grokking [Power et al., 2022] is the delayed rise of validation accuracy long after the training set is fitted, first reported on small algorithmic tasks such as modular arithmetic. Existing explanations stress weight-norm dynamics [Liu et al., 2023], the relative efficiency of memorizing and generalizing circuits [Varma et al., 2023], and a lazy-to-rich transition [Kumar et al., 2024]. Whether and how the delay depends on *model scale* is less clear, yet the answer constrains these accounts and informs the

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

choice of experimental settings: if the delay changed with width, then "the" grokking time of a task would be a property of the model as much as of the task.

We study the width of a one-layer transformer (and of a small MLP) on modular addition, with depth, data and optimizer fixed. Instead of mapping the grid and describing it afterwards, we state a *time-scale account* with three predictions and the results that would refute each (Section 3), and we report which survive. **Contributions.** (i) A pre-specified, censoring-aware analysis of grokking time versus width, with all censored runs kept, an explicit definition of the width below which a run does not fit the training data, and sensitivity to budget, threshold and learning-rate scaling. (ii) An interventional test of a weight-norm mechanism for the width effect (weight decay, initialization scale), which does *not* support it. (iii) A scope map (training fraction, weight decay zero, MLP, multiplication) and an explicit list of what is inconclusive. We do not claim a general law: one prime, 3–5 seeds per cell, widths 8–256.

2 Related work

Grokking was reported on small algorithmic datasets, including modular arithmetic, by Power et al. [2022], with a training-set fraction dependence. Nanda et al. [2023] reverse-engineer a one-layer transformer trained on modular addition and define progress measures that split training into memorization, circuit formation and clean-up; their setup (full-batch AdamW, weight decay 1, one-layer ReLU transformer without LayerNorm) is our reference configuration. Several accounts of the delay exist. Liu et al. [2023] relate it to weight norm and the loss landscape and induce or remove it by changing initialization scale; Varma et al. [2023] explain it by the relative efficiency of memorizing and generalizing circuits; Kumar et al. [2024] tie it to a transition from lazy to rich training dynamics in a two-layer regression setting; Davies et al. [2023] put grokking and double descent in one framework [see also Nakkiran et al., 2020]. Related studies of memorization during training are Tirumala et al. [2022] and Morris et al. [2025], and fitting data without generalizing was examined by Zhang et al. [2017]. Our setting differs from these works in what is varied: width is the controlled variable, with fixed depth, data fraction and optimizer, and the analysis is built around right-censored generalization times instead of discarding runs that do not generalize. We use AdamW [Loshchilov and Hutter, 2017].

3 Method

Task and data. For a prime p we enumerate all p^2 pairs (a, b) with label $(a \circ b) \bmod p$ (addition in the main experiments; subtraction/multiplication where stated). A fraction f of the pairs, chosen by a random permutation seeded per run, is the training set; all remaining pairs are the validation set and are evaluated in full. The main setting is $p = 53$ and $f = 0.4$. No external data are used.

Models and the scale axis. The main model is a one-layer decoder-only transformer on the tokens $[a, b, =]$, read out at the last position, with ReLU MLP, no LayerNorm and untied embeddings, as in Nanda et al. [2023]. *Scale is the model width* d (d_{model}); the number of heads (4, or 1 for $d < 16$) and the MLP hidden size $4d$ follow d, depth is fixed at one layer, and the parameter count is reported per width. Embeddings are initialized $\mathcal{N}(0, 1/d)$. A secondary architecture is an MLP on the concatenated embeddings of a and b with two hidden ReLU layers of size $4d$.

Training. Full-batch AdamW [Loshchilov and Hutter, 2017] (learning rate 10^{-3}, $\beta = (0.9, 0.98)$, weight decay 1 unless varied, 10-step linear warm-up), cross-entropy loss, at most 30,000 steps. In the primary ("fixed") regime all hyperparameters are the same at every width. Train/validation accuracy, loss and the global ℓ_2 weight norm are logged every 250 steps.

Operational definitions (fixed before the main analysis). t_{fit} is the first logged step with training accuracy ≥ 0.99, and t_{gen} the first with validation accuracy ≥ 0.99; the delay is $D = t_{\text{gen}} - t_{\text{fit}}$ (resolution 250 steps). A run without t_{gen} in the budget is *right-censored* at its last step; it is kept and handled by survival methods, never dropped or set to zero.

The account and its pre-stated predictions. We split $t_{\text{gen}} = t_{\text{fit}} + D$. The account says that weight decay must shrink the weight norm from the value at fit time to that of a generalizing solution,

so $D \propto (\mathrm{lr} \cdot \mathrm{wd})^{-1} \log(\|\theta_{\mathrm{fit}}\|/\|\theta^*\|)$, and width acts on D only through t_{fit} and $\|\theta_{\mathrm{fit}}\|$. **A1** (direction): above a minimal width $w_{\min}$ (smallest width at which $\geq 80\%$ of seeds reach training accuracy 0.99), t_{gen} is non-increasing in width; the estimand is the slope of $\log t_{\mathrm{gen}}$ on $\log_2 d$. **A2** (mechanism): (i) $\log D$ has slope -1 in $\log \mathrm{wd}$; (ii) in a regression of $\log D$ on $\log t_{\mathrm{fit}}$ and $\log\|\theta_{\mathrm{fit}}\|$ the coefficient of $\log d$ is zero; (iii) at fixed width, a larger initialization scale (larger norm at fit) lengthens D. **A3** (scope): the training fraction needed for generalization decreases with width, the delay vanishes at large fraction, and nothing generalizes at $\mathrm{wd} = 0$ within the budget. Competing readings are an independent width effect through circuit capacity (C1), a pure initialization-scale/lazy-rich effect (C2), and no width effect beyond noise (C3). A prediction is *supported* if its 95% CI lies wholly on the predicted side, *refuted* if wholly on the other side, and *inconclusive* otherwise; a CI containing a null value is never read as support for it, and unrun cells are never read as null results. No numerical "negligible band" was fixed in advance; we therefore state effect sizes with their CIs.

Amendments made after seeing partial data (post hoc). After the first partial main-sweep logs, the plan replaced "nothing generalizes below $w_{\min}$" by a per-run outcome classification: *grok* (fits, then generalizes), *direct* (generalizes without a separate fitting phase), *memorize* (fits, does not generalize in the budget), *neither*. The step budget was extended for narrow widths after a 100,000-step check, and wd and init-scale cells were added after a feasibility review. Sweeps for a fixed-split control and a second prime were planned and dropped for budget. These changes were made after seeing partial results and are therefore exploratory in nature. The negligibility bands used in Table 2 (A1 slope $|s| < 0.10$; A2(i) tolerance $|b + 1| \leq 0.3$; init-scale slope $|s| < 0.10$) were set after the full results were seen, as round values not tuned to the intervals.

Statistics. Medians of censored times are Kaplan–Meier medians with percentile bootstrap CIs over seeds (2000 resamples); a median can be undefined ("$> x$") when more than half of the runs are censored. Trends use a Jonckheere–Terpstra-type permutation test with pairs ordered only where censoring permits, and a lognormal accelerated-failure-time (AFT) model with right censoring (bootstrap over runs within width, 500 resamples) for slopes. Grokking fractions carry Wilson 95% intervals. A seed fixes both the initialization and the train/validation split, so seeds are independent draws of both.

4 Experiments

All numbers below come from `results/*.csv|json`; Table 1 and the figures are produced by the scripts listed in `paper/REPRODUCE.md`. Each cell has 5 seeds unless noted. The default step ceiling is 30,000.

4.1 A1: wider transformers generalize sooner under fixed hyperparameters

Figure 1 and Table 1 show the main sweep. Training accuracy is reached within 250 steps (the logging resolution) for $d \geq 64$, so there the delay is essentially t_{gen}; narrower nets fit more slowly (t_{fit} median 500 at $d = 32$, 1750 at $d = 16$) and the width-8 net fits in only 1 of 5 runs. Generalization within 30,000 steps occurs in 0/5, 0/5, 1/5, 4/5, 5/5, 4/5 runs at $d = 8, 16, 32, 64, 128, 256$; the KM median of t_{gen} is 18,000, 11,000 and 8,250 steps at $d = 64, 128, 256$ (CIs in Table 1) and undefined below. Over all runs with censoring, t_{gen} decreases with width (JT permutation $p = 0.0002$), and the A1 slope over $d \geq w_{\min} = 16$ is -0.25 per doubling (95% CI $[-0.38, -0.10]$; 25 runs, 11 censored). On the pre-stated decision rule A1 is therefore **supported** (CI wholly below zero; at the 30,000-step budget its upper end -0.10 is at the edge of the post-hoc negligibility band $|\mathrm{slope}| < 0.10$, so only marginally, and clearly at the 100,000-step budget), conditional on fixed learning rate, weight decay and initialization and on widths 16–256; the 30,000-step cells at $d \leq 32$ are lower bounds on t_{gen}, and C3 (no width dependence) is refuted in this regime. At $d = 8$ the picture differs: 4 of 5 runs never reach 99% training accuracy, yet their final validation accuracy is 0.56–0.97, so these runs show substantial validation accuracy without a clean fit-then-generalize pattern (classified "neither" in Table 1); this width is not a grokking case and is excluded from the slope. Three qualifications matter. (a) *The 30,000-step ceiling censors the narrow widths.* In a 100,000-step check with 3 seeds at $d = 16$ and 32 (which reproduce the 30,000-step runs exactly), 1/3 and 2/3 runs generalize, the three that generalize doing so between 35,000 and 58,000 steps (KM median at $d = 32$: 38,500). So with a longer budget some narrow runs generalize late, which shows that the 30,000-step cells

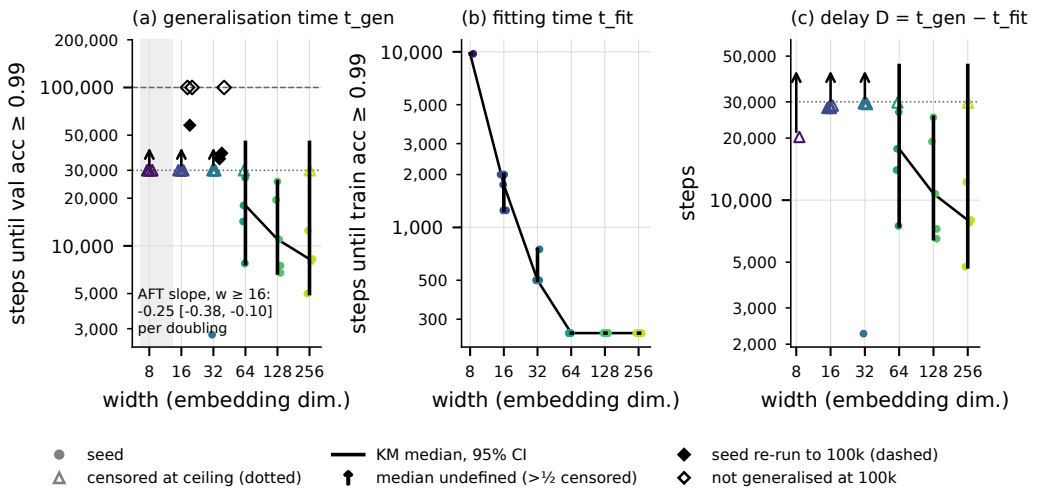


Figure 1: Timing versus width (one-layer transformer, addition mod 53, training fraction 0.4, AdamW, learning rate 10^{-3}, weight decay 1; 5 seeds). (a) t_{gen} over all runs; censored runs are open triangles at the ceiling and are kept. (b) t_{fit} and (c) the delay exist only for runs that fit. Black: Kaplan–Meier median with 95% bootstrap CI; arrow: median undefined. Shaded: widths below $w_{\min}$, excluded from the AFT slope printed below (a). Diamonds in (a): seeds 0–2 at $d = 16, 32$ re-run to a 100,000-step ceiling; the 30,000-step value there is a lower bound.

Table 1: Main sweep per width (5 seeds each). Regimes G/D/M/N: seeds that grok / generalise directly / memorise only / neither. Medians are Kaplan–Meier with 95% bootstrap CIs over seeds; $> x$: median undefined (over half the seeds censored at 30,000 steps), x = smallest observed value; ∞: CI bound undefined. Censored runs are never dropped.

width	params	grokked	regimes G/D/M/N	KM median t_{gen} [95% CI]	median t_{fit} [95% CI]	KM median delay [95% CI]
8	1688	0/5	0/0/1/4	>30,000	9,750 [9,750, 9,750]	>20,250
16	4912	0/5	0/0/5/0	>30,000	1,750 [1,250, 2,000]	>28,000
32	15968	1/5	1/0/4/0	>2,750	500 [500, 750]	>2,250
64	56512	4/5	4/0/1/0	18,000 [7,750, ∞]	250 [250, 250]	17,750 [7,500, ∞]
128	211328	5/5	5/0/0/0	11,000 [6,750, 25,500]	250 [250, 250]	10,750 [6,500, 25,250]
256	815872	4/5	4/0/1/0	8,250 [5,000, ∞]	250 [250, 250]	8,000 [4,750, ∞]

are censored; with 3 seeds we cannot say what fraction of narrow runs ever generalize (Figure 8). With these runs the slope is -0.41 $[-0.57, -0.25]$, but restricted to $d \geq 32$ it is -0.31 $[-0.62, 0.12]$, which includes zero: *inconclusive* on that subrange. (b) *Threshold.* The slope is stable for the 0.9, 0.95 and 1.0 validation thresholds (-0.38, -0.36, -0.24; the 1.0 analysis is censored at 22/25 runs because runs stop 500 steps after the 0.99 crossing, so it is a bound on timing and weak: its interval $[-0.37, -0.10]$ reaches the band edge and is *inconclusive*; the 0.9 and 0.95 analyses are exact and supported). We therefore re-ran the 13 grokked runs at $d \geq 64$ (bit-identical to the originals up to t_{gen}) for 5,000 steps past the 0.99 crossing: 9 of 13 reach validation accuracy 1.0 (median lag 2,250 steps), and the exact 1.0 slope over $d \geq 64$ is -0.28 $[-0.55, 0.02]$, against -0.26 $[-0.59, 0.09]$ for the 0.99 threshold on the same runs: the same estimate, inconclusive for both because only three widths are in the window. (c) *It is not monotone in the grok fraction*: 4/5 at $d = 64$, 5/5 at 128, 4/5 at 256; with five seeds this is within noise. The same direction appears for the MLP (JT $p = 0.0002$, AFT slope -0.38 $[-0.40, -0.36]$); this CI is narrow because width 16 is fully censored at the ceiling and the AFT likelihood leans on the lognormal assumption, so we read it as direction only. Figure 2 shows the fit-then-generalize shape; survival curves and regime counts are in Appendix A.

4.2 A2: the weight-norm mechanism is not supported; only the weight-decay scaling survives

(i) Weight decay. Varying weight decay over $\{0.5, 1, 2\}$ at fixed learning rate, the slope of $\log D$ on $\log \text{wd}$ is -1.01 $[-1.38, -0.66]$ at $d = 128$ (consistent with -1) and -0.61 $[-1.05, -0.12]$ at $d = 64$ (also consistent with -1, but wide and not close to it in its point estimate; Figure 3a). At

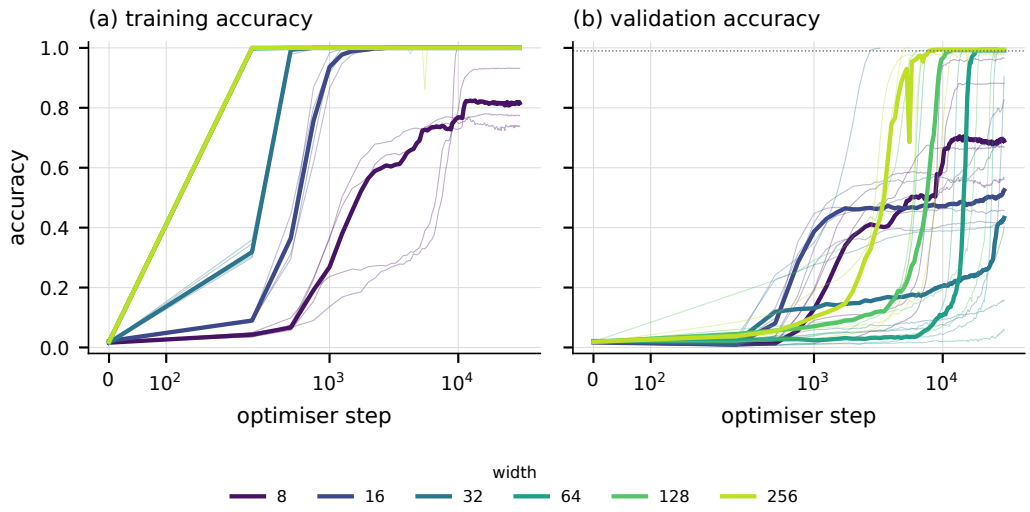


Figure 2: Training (a) and validation (b) accuracy during training for the main sweep (thin: single seeds, thick: median per width; logarithmic step axis).

$d = 64$ the dependence is not monotone: the median delay is 17,750 at wd=1 and 23,500 at wd=2, so the fitted slope rests on the wd=0.5 cell, which has a 60,000-step ceiling. Against the tolerance $|b + 1| \leq 0.3$ neither interval lies wholly inside $[-1.3, -0.7]$, so we call the exponent of A2(i) *inconclusive*; the direction (slope < 0) is supported at both widths and not refuted. **(ii) Mediation by norm.** We regress $\log D \sim \log t_{\text{fit}} + \log\|\theta_{\text{fit}}\| + \log d$ over all 66 transformer runs with $f = 0.4$ (24 censored) and read "width acts only through the norm" as refuted if the width coefficient's interval excludes zero. The answer depends on the norm measure. With the *total* L_2 norm the width coefficient is -0.57 $[-1.17, -0.13]$ ("refuted"), but the total norm grows with $\sqrt{\#\text{parameters}}$, hence about linearly in width, so it is collinear with width by construction (it rises from 23.5 at $d = 8$ to 49.3 at $d = 256$ while the delay shrinks). With the *per-parameter RMS* norm ($L_2/\sqrt{\#\text{parameters}}$) the width coefficient is $+0.59$ $[-0.83, 2.26]$: *inconclusive*; the norm coefficient without width is 0.52 $[0.13, 0.89]$ (positive as predicted, but not beyond the negligibility band, inconclusive), against -0.95 $[-2.18, 0.56]$ for the total norm. On the 26 main-sweep runs (five widths) the width coefficients have opposite signs under the two measures (-1.70 $[-4.26, -0.85]$ and $+3.68$ $[0.34, 12.6]$), which shows that the regression cannot separate width from norm there. We therefore call A2(ii) *inconclusive (refuted only under the total L_2 norm)*. The norm at t_{gen} is contaminated by the outcome and is not used. **(iii) Initialization scale.** Multiplying all initial parameters by $\alpha \in \{0.5, 1, 2\}$ raises the norm at fit (at $d = 128$: 39.1, 42.8, 48.3), yet shortens generalization: the slope of $\log t_{\text{gen}}$ on $\log \alpha$ is -0.57 $[-0.95, -0.08]$ at $d = 128$ and -0.18 $[-0.44, -0.01]$ at $d = 32$ (borderline; Figure 3b). This is a *total* effect of initialization scale. At $d = 128$ the norm at fit rises with α while t_{gen} falls, a direction opposite to "larger norm $\Rightarrow$ longer delay" (direction-only: the interval is wholly below zero); under the strict band reading the verdict is *inconclusive*, since the interval's upper end, -0.08, lies above the band edge -0.10. At $d = 32$ the norm at fit is not monotone in α (34.0, 32.6, 34.8), so it is *not tested* there. A norm-matched comparison was not achievable (the lowest norm at $d = 128$, 39.1, still exceeds that of $d = 32$ at $\alpha = 1$, 32.6; $\alpha < 0.5$ was not run), so mediation through the norm at fit remains untested by intervention. Overall the account's norm-mediation mechanism is *not supported*: no test supports it and two point against it, neither decisively (ii, iii); only the weight-decay scaling (i) survives, as a time scale not shown to run through the norm. The data do not separate an independent capacity effect (C1) from an initialization-scale-only effect (C2); and since the lr-rescaling check (3–5 seeds, mostly censored) cannot exclude that part of the width advantage depends on the fixed-learning-rate regime, C1 is at most one compatible reading.

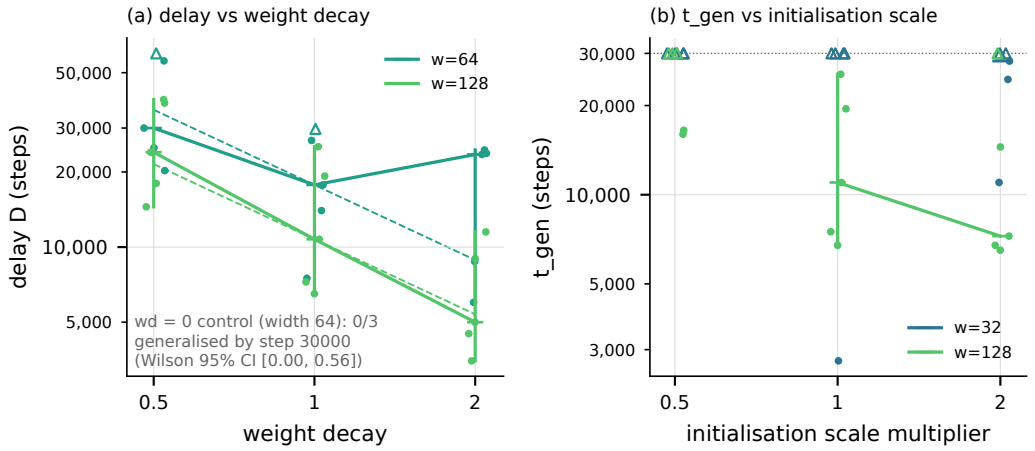


Figure 3: (a) Delay versus weight decay at widths 64 and 128 (dashed: AFT fit; slope -1 is the prediction); the wd=0 control (width 64) did not generalize in 0/3 runs. (b) t_{gen} versus initialization-scale multiplier at widths 32 and 128. Markers as in Figure 1.

4.3 A3 and scope: training fraction, parametrization, other tasks

Training fraction (Figure 4). At $f = 0.25$ none of 5+5 runs at $d = 32$ or 128 generalize (mean final validation accuracy 0.02 and 0.10); at $f = 0.4$, $1/5$ runs generalize at $d = 32$ (Wilson $[0.04, 0.62]$) versus $5/5$ at $d = 128$ ($[0.57, 1.0]$), the pre-stated pattern "5/5 versus $\leq 1/5$" holds, but with five seeds the Wilson intervals overlap, so this contrast alone is inconclusive; at $f = 0.5$ all runs generalize at both widths, and the wider net is faster (KM median t_{gen} 2,750 vs 9,000). The width contrast in the fraction that generalizes is thus seen at $f = 0.4$ only, not at 0.25 or 0.5, so a critical-fraction shift with width is *inconclusive*; by the Wilson rule, the supported contrasts are $f = 0.5$ versus 0.25 ($5/5$ versus $0/5$ at both widths) and $f = 0.4$ versus 0.25 at $d = 128$, and at $f = 0.5$ width still shortens t_{gen}. A finer fraction grid was not run. The delay does not vanish at $f = 0.5$ (2,500 steps at $d = 128$), so "delay vanishes at large f" is untested within our range. **Weight decay zero** (width 64, 3 seeds): 0/3 generalize in 30,000 steps (Wilson upper bound 0.56); this is consistent with A3 but has low power with three seeds, and is not a strong test. **Parametrization** (learning rate multiplied by $\sqrt{64/d}$; 5 seeds at $d \in \{32, 128\}$, 3 seeds at $d \in \{16, 64, 256\}$). At $d = 32$ and 128, $2/5$ and $3/5$ runs generalize (fixed rate: $1/5$ and $5/5$); the width contrast has Fisher $p = 1.0$ in the scaled arm and 0.048 at fixed rate. Over $d = 16, \ldots, 256$ (3 seeds each) $0, 0, 3, 1, 1$ of 3 generalize against $0, 0, 3, 3, 2$ at fixed rate. The scaled-rate AFT slope is degenerate (10 of 15 runs censored, its bootstrap interval collapses to $[-0.05, 0.00]$) and inconclusive, whereas the fixed-rate slope on the same 15 runs is -0.20 $[-0.32, -0.07]$. The width advantage at $d \geq 128$ is thus weaker when the learning rate is scaled with width, but 30,000 steps censor most scaled-rate runs, so we can neither exclude nor confirm a dependence on the learning rate; all conclusions are stated *under fixed learning rate, weight decay and initialization*. **Other settings.** Two-hidden-layer MLPs show the same direction with faster generalization (KM median t_{gen} 13,000, 8,000, 6,750, 6,000 at $d = 32, 64, 128, 256$; Table 3, Figure 9). Modular multiplication (2 widths $\times$ 3 seeds, $d = 64, 128$): all 6 runs generalize, KM median 22,500 versus 19,000 steps, CIs overlapping.

4.4 Summary of verdicts

5 Discussion and limitations

Under fixed learning rate, weight decay and initialization, a wider transformer generalizes sooner on this task, and the delay's inverse dependence on weight decay is consistent with a weight-decay-driven cleanup time scale. The weight-norm mechanism we proposed for the width effect is not supported: whether width keeps a coefficient after conditioning on the norm at fit depends on the norm measure (yes for the total L_2 norm, which is collinear with width; inconclusive for the per-parameter RMS norm), and at $d = 128$ the intervention that raises the norm at fit (larger initial scale) shortens the

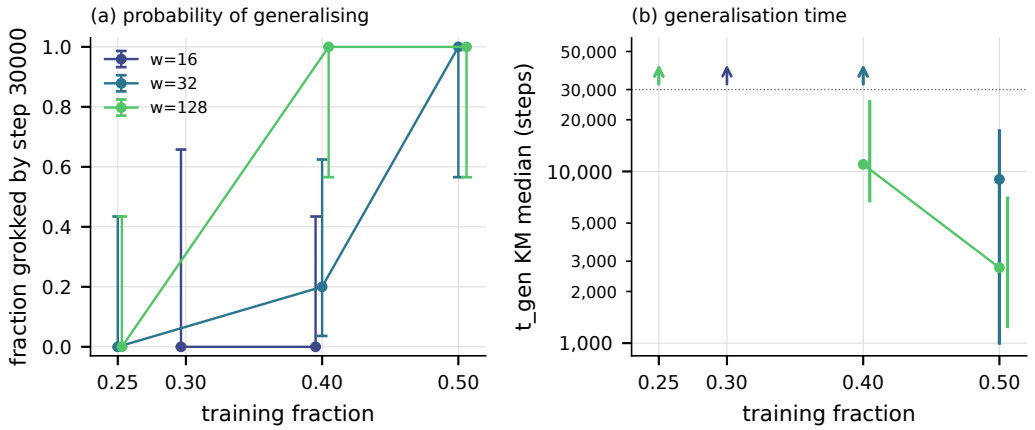


Figure 4: (a) Fraction of seeds generalizing within 30,000 steps versus training fraction, with Wilson 95% intervals ($n = 5$ per point, 2 at $f = 0.3$, $d = 16$). (b) KM median t_{gen} versus fraction (arrow: undefined).

Table 2: Pre-stated predictions and outcomes under the decision rule of Section 3. Evidence: `results/main_stats.json`, `sens_stats.json`, `mech_stats.json`, `*_cells.csv`.

Prediction	Outcome	Basis
A1 wider $\Rightarrow$ earlier t_{gen} ($d \geq 16$, fixed lr/wd/init)	supported	slope -0.25 $[-0.38, -0.10]$ (30k); -0.41 $[-0.57, -0.25]$ (100k, narrow widths); MLP -0.38; inconclusive on $d \geq 32$ alone (-0.31 $[-0.62, 0.12]$)
A2(i) $D \propto 1/\text{wd}$	direction supported; exponent inconclusive	-1.01 $[-1.38, -0.66]$ ($d = 128$); -0.61 $[-1.05, -0.12]$ ($d = 64$); neither inside $[-1.3, -0.7]$
A2(ii) width acts only via t_{fit} and norm at fit	inconclusive (refuted only under total L_2)	width coefficient -0.57 $[-1.17, -0.13]$ (total L_2); $+0.59$ $[-0.83, 2.26]$ (per-parameter RMS); the two norms are collinear with width differently
A2(iii) init scale / norm matching	direction-only against ($d = 128$); verdict inconclusive; not tested at $d = 32$	$\log t_{\text{gen}}$ slope in $\log \alpha$ -0.57 $[-0.95, -0.08]$ ($d = 128$), norm rises with α; norm matching not achievable
A2 mechanism overall	not supported	no test supports it; two point against, neither decisive
A3 critical fraction shifts with width	inconclusive	1/5 vs 5/5 at $f = 0.4$; 0/5 vs 0/5 at 0.25, 5/5 vs 5/5 at 0.5
A3 no grokking at wd=0	consistent, low power	0/3 (Wilson upper bound 0.56)
A3 delay vanishes at large f	untested	max $f = 0.5$
Parametrization (lr $\propto 1/\sqrt{d}$)	open (flag)	$d = 128$ vs 32: 3/5 vs 2/5 (scaled), 5/5 vs 1/5 (fixed), Fisher $p = 1.0$ vs 0.048; scaled AFT slope inconclusive (10/15 censored)
C3 no width effect	refuted (fixed regime)	A1 slope, JT test
C1 vs C2	not separable; C1 one compatible reading	A2(ii), A2(iii), lr-rescaling

delay; a norm-matched test was not feasible. The initial-scale result is a total effect, compatible with larger initial scale favoring richer feature learning [Kumar et al., 2024, Liu et al., 2023], but we did not test this, and we cannot separate an independent capacity effect from an initialization-scale effect.

Limitations and failure cases. (1) One task family and prime ($p = 53$), mostly addition; multiplication with 3 seeds and 2 widths. (2) Width range 8–256 at one layer; depth and the number of heads were not varied (heads change from 1 to 4 between $d = 8$ and $d \geq 16$). (3) Five seeds per cell (3 for several controls); intervals are wide, and several verdicts are "inconclusive" or "narrow". The $d = 16$ and $d = 32$ medians are undefined at the 30,000-step ceiling. (4) Hyperparameters are fixed across width; the learning-rate-scaled check weakens the width advantage at $d \geq 128$ but is mostly censored at 30,000 steps (3–5 seeds), so it neither excludes nor confirms a dependence, and we did not tune each width. (5) A1 uses a lognormal AFT model; its CIs are bootstrap intervals conditional on that model. The logging interval of 250 steps bounds the resolution of t_{fit} (which is at the resolution for $d \geq 64$). (6) Not run: the split-seed versus init-seed separation, a second prime, the Nanda-configuration anchor ($p = 113$), $\alpha < 0.5$, a finer training-fraction grid, depth, optimizer and batch-size variation, and a norm-matched comparison; the exact 1.0-threshold check covers only $d \geq 64$ (inconclusive), and 0.9/0.95 thresholds are exact. The 30,000-step cells at narrow widths are lower bounds on t_{gen}. (7) The plan was amended after partial data were seen (Section 3); the A2 regression is collinear in width and norm, the negligibility bands were fixed after the results were seen, and the post-hoc outcome classification is exploratory. (8) We did not use mechanistic diagnostics (Fourier or restricted-loss analyses), so claims about circuits are interpretations, not findings.

Reproducibility. Code, run logs and the artifact map are in the delivered repository (`paper/REPRODUCE.md`); data are generated in memory.

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A Additional figures and tables

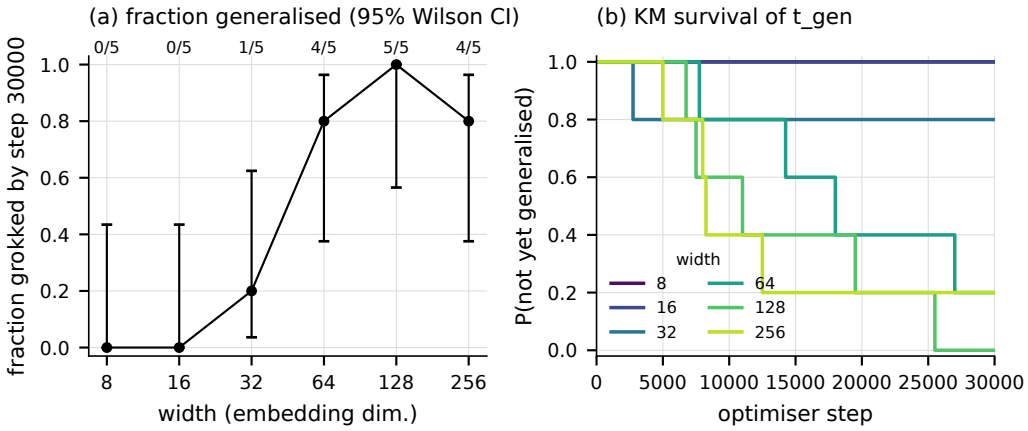


Figure 5: Probability of generalizing and survival of t_{gen} by width.

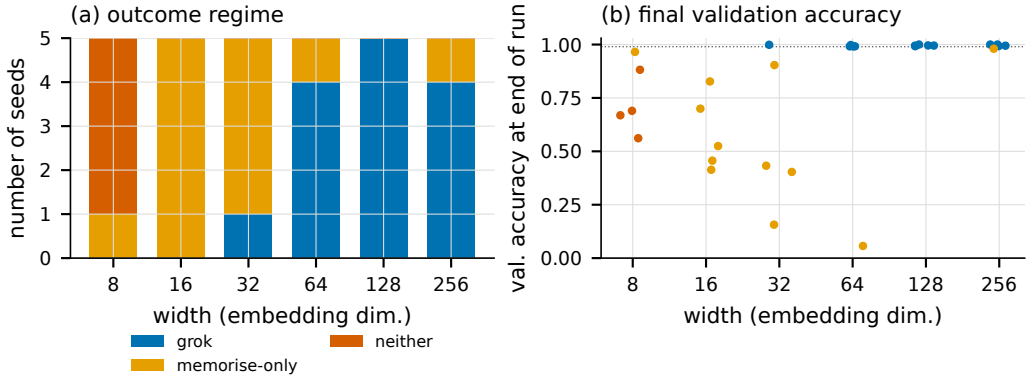


Figure 6: Outcome regimes per width.

B Use of LLM agents

The study (plan, code, runs, analysis and manuscript) was carried out by LLM-based agents working as separate roles (planning, experiments, figures, venue, writing, review) under a human-specified task. They are not part of the studied method. All numbers in this paper come from the shipped run logs and analysis scripts; the agents' roles and hand-offs are recorded in the delivery repository's issue and pull-request history.

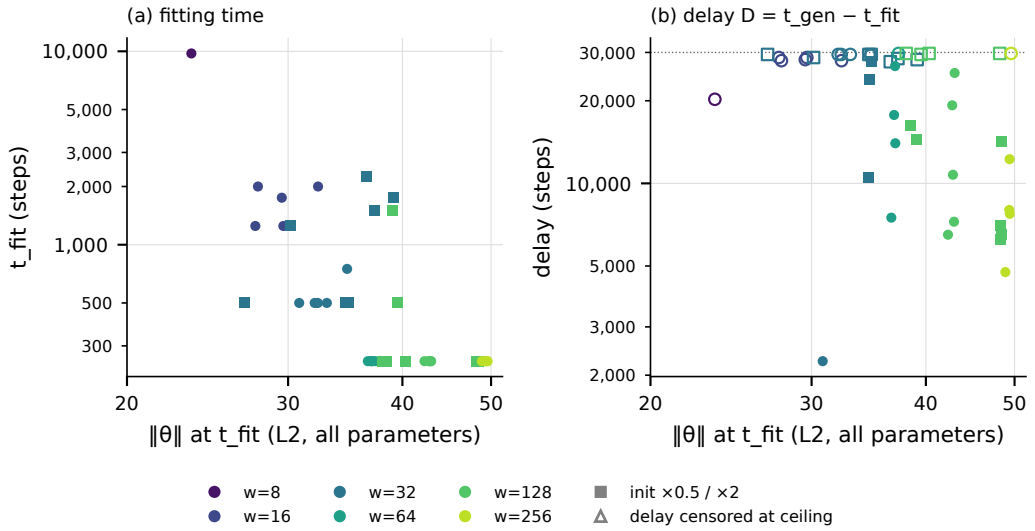


Figure 7: Weight norm at the time the training set is fitted versus (a) t_{fit} and (b) the delay, one point per run (colour: width). Circles: default initialization; squares: initialization scaled by 0.5 or 2. Open markers in (b): runs not generalized within the ceiling. Only runs that fitted the training set appear. Norm is the total L_2 norm.

Table 3: Scope sweeps: same columns as Table 1 for secondary conditions.

sweep	condition	width	grokked	regimes G/D/M/N	KM median t_{gen} [95% CI]	median t_{fit} [95% CI]	KM median delay [95% CI]
frac	train frac 0.25	32	0/5	0/0/5/0	>30,000	500 [500, 500]	>29,500
frac	train frac 0.25	128	0/5	0/0/5/0	>30,000	250 [250, 250]	>29,750
frac	train frac 0.3	16	0/2	0/0/2/0	>30,000	3,125 [1,250, 5,000]	>25,000
frac	train frac 0.5	32	5/5	5/0/0/0	9,000 [1,000, 17,250]	750 [500, 1,250]	8,250 [500, 16,000]
frac	train frac 0.5	128	5/5	5/0/0/0	2,750 [1,250, 7,000]	250 [250, 250]	2,500 [1,000, 6,750]
wd	wd 0.5	64	4/5	4/0/1/0	30,250 [20,500, ∞]	250 [250, 250]	30,000 [20,250, ∞]
wd	wd 0.5	128	5/5	5/0/0/0	24,250 [14,750, 39,250]	250 [250, 250]	24,000 [14,500, 39,000]
wd	wd 2	64	5/5	5/0/0/0	24,000 [6,250, 25,000]	500 [250, 500]	23,500 [6,000, 24,500]
wd	wd 2	128	5/5	5/0/0/0	5,250 [3,750, 11,750]	250 [250, 250]	5,000 [3,500, 11,500]
wd0	wd 0	64	0/3	0/0/3/0	>30,000	250 [250, 250]	>29,750
alpha	init ×0.5	32	0/5	0/0/5/0	>30,000	1,500 [500, 2,250]	>27,750
alpha	init ×2	32	3/5	3/0/2/0	28,250 [11,000, ∞]	500 [500, 500]	27,750 [10,500, ∞]
alpha	init ×0.5	128	2/5	2/0/3/0	>16,000	250 [250, 1,500]	>14,500
alpha	init ×2	128	4/5	4/0/1/0	7,250 [6,500, ∞]	250 [250, 250]	7,000 [6,250, ∞]
lrscale	lr $\propto 1/\sqrt{w}$ (ref 64)	32	2/5	2/0/3/0	>5,500	500 [500, 500]	>5,000
lrscale	lr $\propto 1/\sqrt{w}$ (ref 64)	128	3/5	3/0/2/0	25,000 [15,000, ∞]	250 [250, 250]	24,750 [14,750, ∞]
lrscale2	lr $\propto 1/\sqrt{w}$ (ref 64)	16	0/3	0/0/3/0	>30,000	1,000 [1,000, 1,000]	>29,000
lrscale2	lr $\propto 1/\sqrt{w}$ (ref 64)	64	3/3	3/0/0/0	14,250 [7,750, 27,000]	250 [250, 250]	14,000 [7,500, 26,750]
lrscale2	lr $\propto 1/\sqrt{w}$ (ref 64)	256	1/3	1/0/2/0	>29,750	250 [250, 250]	>29,500
mlp	arch mlp	8	1/5	1/0/1/3	>10,750	3,750 [2,000, 5,500]	8,750 [8,750, ∞]
mlp	arch mlp	16	0/5	0/0/5/0	>30,000	750 [750, 750]	>29,250
mlp	arch mlp	32	5/5	5/0/0/0	13,000 [12,250, 14,750]	250 [250, 250]	12,750 [12,000, 14,500]
mlp	arch mlp	64	5/5	5/0/0/0	8,000 [4,750, 10,000]	250 [250, 250]	7,750 [4,500, 9,750]
mlp	arch mlp	128	5/5	5/0/0/0	6,750 [6,250, 9,500]	250 [250, 250]	6,500 [6,000, 9,250]
mlp	arch mlp	256	5/5	5/0/0/0	6,000 [5,500, 7,250]	250 [250, 250]	5,750 [5,250, 7,000]
mul	op mul	64	3/3	3/0/0/0	22,500 [16,750, 28,000]	500 [500, 500]	22,000 [16,250, 27,500]
mul	op mul	128	3/3	3/0/0/0	19,000 [4,000, 24,250]	250 [250, 250]	18,750 [3,750, 24,000]

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and Section 1 state the task, width range, single prime, and that outcomes are supported, refuted or inconclusive; Table 2 lists each.

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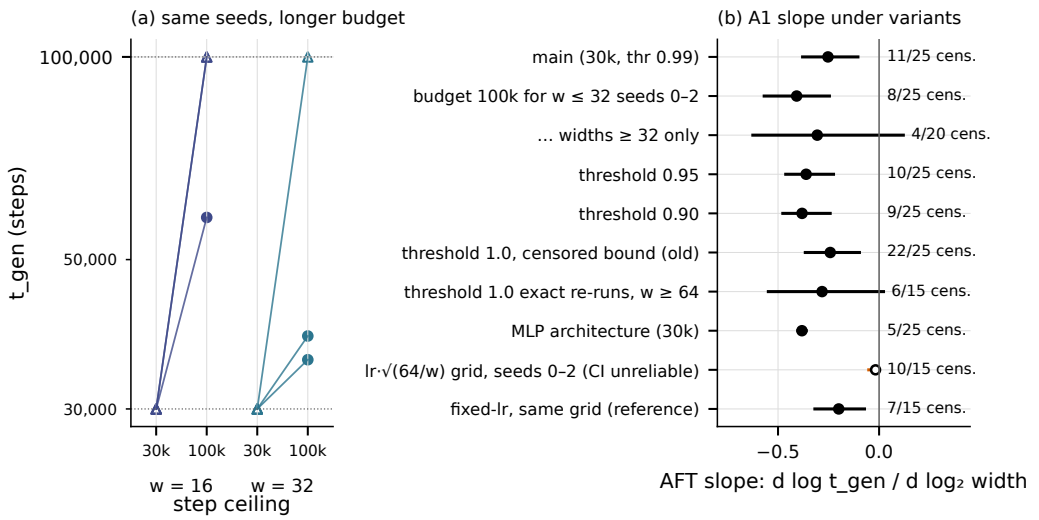


Figure 8: Sensitivity of the width effect to step budget, threshold and architecture. (a) Seeds 0–2 at widths 16 and 32 under the 30,000-step ceiling (main) and a 100,000-step ceiling (long; identical through step 30,000); open triangles: not generalized at that ceiling. (b) Censoring-aware lognormal AFT slope of $\log t_{\text{gen}}$ on $\log_2 d$ over $d \geq w_{\min}$ with 95% bootstrap CIs over runs (negative: wider generalizes sooner); right: censored/total runs. The first threshold-1.0 row is a censored bound only; the exact row re-ran widths ≥ 64 to validation accuracy 1.0. The learning-rate-scaled row is the AFT slope over widths 16–256 (seeds 0–2) with 10 of 15 runs censored, so its interval collapses toward 0 and is not evidence of no effect; the row beneath is the fixed-rate slope on the same runs.

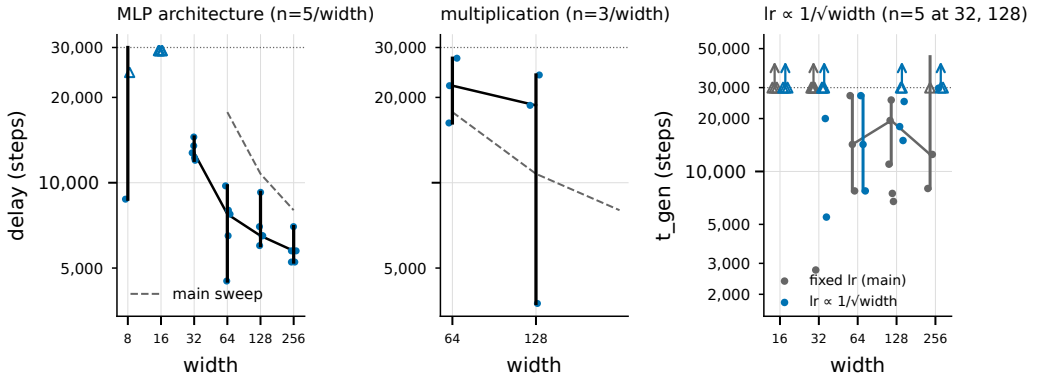


Figure 9: Secondary conditions. (a, b) Delay versus width for the MLP and for multiplication, shared y-axis. Filled: generalized; open triangle: right-censored at the ceiling. Black: Kaplan–Meier median with 95% bootstrap CI; dashed grey: main-sweep medians. (c) Learning rate scaled as $\text{lr}\sqrt{64/d}$ (blue) against fixed-rate main-sweep runs on the same seeds (grey), t_{gen} on the y-axis because most scaled-rate runs do not generalize. Points: seeds 0–4 at $d = 32, 128$ and seeds 0–2 at $d = 16, 64, 256$; lines: KM median over seeds 0–2.

- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
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- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

Table 4: Trend of generalisation time with width. JT: Jonckheere–Terpstra-type statistic, negative = faster for wider nets, two-sided permutation p, censoring-aware. AFT: lognormal accelerated-failure-time slope of $\log t_{gen}$ on $\log_2$ width, 95% bootstrap CI over seeds (the plan's A1 estimand; negative = wider nets generalise sooner). Verdict: the plan's rule with the negligible bands of story-repo's `VERDICTS.md`, as written by the analysis position into the stats files (bands set after the results were seen). The scaled-learning-rate row is censored for 10 of 15 runs; with most runs sharing one censoring time the bootstrap CI collapses toward 0, so an interval inside the negligible band is not evidence of no effect (the verdict is inconclusive, not null).

sweep	condition	endpoint	estimate	test	verdict
main	transformer, add mod 53, f=0.4	t_gen, all runs	JT = -199	permutation p = 0.0002	–
main	transformer, add mod 53, f=0.4	log t_gen per doubling (AFT, w $\geq$ 16)	-0.25 [-0.38, -0.10]	n = 25 (11 censored)	supported
main	transformer, add mod 53, f=0.4	delay, runs that fit	JT = -141	permutation p = 0.0006	–
mlp	MLP, add mod 53, f=0.4	t_gen, all runs	JT = -296	permutation p = 0.0002	–
mlp	MLP, add mod 53, f=0.4	log t_gen per doubling (AFT, w $\geq$ 16)	-0.38 [-0.40, -0.36]	n = 25 (5 censored)	supported
mlp	MLP, add mod 53, f=0.4	delay, runs that fit	JT = -232	permutation p = 0.0002	–
sensitivity	100k-step ceiling for w $\leq$ 32, seeds 0–2	log t_gen per doubling (AFT, w $\geq$ 16)	-0.41 [-0.57, -0.25]	n = 25 (8 censored)	supported
sensitivity	same, widths $\geq$ 32 only	log t_gen per doubling (AFT, w $\geq$ 32)	-0.31 [-0.62, 0.12]	n = 20 (4 censored)	inconclusive
sensitivity	threshold 1.0 (bound only)	log t_gen per doubling (AFT, w $\geq$ 16)	-0.24 [-0.37, -0.10]	n = 25 (22 censored)	inconclusive
sensitivity	threshold 0.90	log t_gen per doubling (AFT, w $\geq$ 16)	-0.38 [-0.48, -0.24]	n = 25 (9 censored)	supported
sensitivity	threshold 0.95	log t_gen per doubling (AFT, w $\geq$ 16)	-0.36 [-0.46, -0.23]	n = 25 (10 censored)	supported
sensitivity	threshold 1.0, exact re-runs (thr1)	log t_gen per doubling (AFT, w $\geq$ 64)	-0.28 [-0.55, 0.02]	n = 15 (6 censored)	inconclusive
sensitivity	lr $\propto 1/\sqrt{w}$, seeds 0–2, w 16–256	log t_gen per doubling (AFT, w $\geq$ 16); CI unreliable	-0.02 [-0.05, -0.00]	n = 15 (10 censored)	inconclusive
sensitivity	fixed lr, same seeds and widths (reference)	log t_gen per doubling (AFT, w $\geq$ 16)	-0.20 [-0.32, -0.07]	n = 15 (7 censored)	–

Table 5: A2(ii): regression of $\log D$ (lognormal AFT, censored runs kept) on $\log t_{fit}$ and the weight norm at t_{fit} ($\pm$ log width), with the norm measured as the total L2 norm of all parameters and as the per-parameter RMS (total/$\sqrt{\text{params}}$; total L2 grows with width by construction). Cells: coefficient [95% bootstrap CI] and the verdict under the plan's bands (norm coefficient predicted > 0: supported if the CI lies above 0.25, refuted if below 0.25). For the width row the verdict is for "width acts only through t_{fit} and the norm" (refuted if the width CI excludes 0). Width and norm are collinear, so these are supporting only. Regressions on the norm at t_{gen} are outcome-contaminated and not shown.

sample	coefficient	total L2 [CI]	total L2 verdict	RMS [CI]	RMS verdict
main sweep (n=26, 12 cens.)	log norm (given $\log t_{fit}$)	-0.96 [-3.59, 1.55]	inconclusive	0.51 [-0.13, 1.02]	inconclusive
main sweep (n=26, 12 cens.)	log norm (given $\log t_{fit}$, log width)	6.73 [1.40, 19.84]	supported	5.26 [1.26, 16.84]	supported
main sweep (n=26, 12 cens.)	log width (given $\log t_{fit}$, log norm)	-1.70 [-4.26, -0.85]	refuted	3.68 [0.34, 12.65]	refuted
all f=0.4 runs (n=66, 24 cens.)	log norm (given $\log t_{fit}$)	-0.95 [-2.18, 0.56]	inconclusive	0.52 [0.13, 0.89]	inconclusive
all f=0.4 runs (n=66, 24 cens.)	log norm (given $\log t_{fit}$, log width)	1.04 [-0.94, 3.74]	inconclusive	1.28 [-0.50, 3.47]	inconclusive
all f=0.4 runs (n=66, 24 cens.)	log width (given $\log t_{fit}$, log norm)	-0.57 [-1.17, -0.13]	refuted	0.59 [-0.83, 2.26]	inconclusive

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 lists limitations (one task family and prime, narrow width range, few seeds, fixed hyperparameters, censoring, post-hoc plan amendments, no mechanistic analysis).

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Table 6: A2(i) weight-decay exponent and A2(iii) init-scale effect (lognormal AFT, censored runs kept; the $wd = 0.5$ runs have a 60,000-step ceiling). A2(i): supported if the CI lies inside $[-1.3, -0.7]$, refuted if wholly outside. A2(iii): the mediation account predicts a positive slope where the norm at fit rises with α (supported if lo > 0.10, refuted if hi < -0.10; "not tested" where the norm at fit is not monotone in α); the direction-only column asks only whether hi < 0.

estimand	setting	n (censored)	estimate [CI]	verdict	direction-only
A2(i): $d \log D / d \log wd$ (predicted -1)	width 64, wd $\in$ 0.5, 1, 2	15 (2)	-0.61 [-1.05, -0.12]	inconclusive	–
A2(i): $d \log D / d \log wd$ (predicted -1)	width 128, wd $\in$ 0.5, 1, 2	15 (0)	-1.01 [-1.38, -0.66]	inconclusive	–
A2(iii): $d \log t_{gen} / d \log \alpha$ (init scale)	width 32, $\alpha \in$0.5, 1, 2	15 (11)	-0.18 [-0.44, -0.01]	not tested	not tested
A2(iii): $d \log t_{gen} / d \log \alpha$ (init scale)	width 128, $\alpha \in$0.5, 1, 2	15 (4)	-0.57 [-0.95, -0.08]	inconclusive	refuted

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
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Justification: Section 3 (split rule: random train/validation split of all p^2 pairs per seed; no hyperparameter search, hyperparameters fixed a priori and varied only as stated) and `REPRODUCE.md`.

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